

Jethro Au

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Research Interest

Focused on explainability, safety, interpretability, and scientific grounding of large-scale ML systems, with experience in generative modeling, multi-modality, XAI, and clinical foundational models. Motivated to advance reliable and steerable AI system for scientific reasoning and automated scientific discovery.

Education

Harvard University — MS Health Data Science (Aug 2025 – Mar 2027)

Area of Focus: Multi-Modal LLMs, Explainable Clinical Agents, AI for Health

Courses: Computational Optimal Transport for ML, Computing for Big Data, Statistical Learning, Epidemiology

Massachusetts Institute of Technology — Cross Registration (Jan 2026 – Mar 2027)

Courses: Modeling in Machine Learning, Machine Learning for Molecular Engineering, Advances in Computer Vision

Hong Kong University of Science and Technology — MS Big Data Technology TPG Fellow (Sept 2023 – Jun 2025)

TPG Fellow | HKUST School of Engineering (CS & Math)

Research Focus: ML Research in AlphaFold, LLM, Parallel Programming, Social Computing (Graph Networks), CV

Northwestern University — B.S. Industrial Engineering

McCormick School of Engineering, Certificate in Managerial Analytics - Kellogg School of Management

Courses: Stats & Machine Learning, Optimization Algorithms, Stochastic Simulation, Calculus, Data Management

Technical Experience

Machine Learning Engineer Intern - Apple (AIML Health) (May 2026 - Aug 2026)

- Researched, developed, and trained an end-to-end architecture utilizing complex, heterogeneous public knowledge graphs to enhance explainable and steerable model reasoning over EHR data.
- Evaluated system performance against medical reasoning benchmarks, demonstrating successful improvements in complex domain-specific tasks.
- Developed and deployed internal tooling and plugins to scale advanced reasoning capabilities.
- Delivered technical seminars and demos to the Health AI team.

Research Intern - AI & Robotics Lab, Harvard Medical School (Sept 2025 - May 2026)

- Led research project team on text-to-genomic multimodality models including: genomic-text multimodality alignment and text-guided genomic sequencing design.
- Conducted post-training SFT and contrastive learning with DNA foundational models (AlphaGenome & Evo2) with GenePT embeddings for textual steerability in DNA-sequence phenotyping predictions.

AI/ML Engineer (Summer) — Apricot AI, Hong Kong (Jun 2025 – Aug 2025)

- Fine-tuned clinical LLM via distillation, improving clinical diagnosis, reasoning, and treatment recommendations from doctor-patient transcripts.
- Improved clinical reasoning of model via GRPO preference alignment with curriculum learning and direct-RLAIF, based on Google Deepmind's ICML 2024 work.

ML Research – Generative Modeling for Protein Design — HKUST (Jul 2024 - Jul 2025)

- Researched AlphaFold-driven de novo cyclic peptide design for HIV and Hepatitis-C under Prof. Nevin Zhang and Dr. Tao Wang (Rukobia inventor).
- Developed candidate peptides with a backpropagation-based steering algorithm, improving HIV CD4-binding precision of generated cyclic peptides.
- Optimized AlphaFold2 AfCycDesign, validated against molecular simulation benchmarks.

Project Lead (Data & Analytics) – COVID-19 RT-PCR Screening — Prenetics, Hong Kong (Jun 2021 – May 2023)

- Improved reporting accuracy on 2M+ COVID tests, reducing processing errors by 30% and scaling on-site throughput to 400 staff per day.
- Built ETL + ML pipelines to improve reporting accuracy, improved on-site efficiency via human-centric design.

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Selected ML Work

(Ongoing) Multimodal Clinical Discordance: Capstone — MIT Laboratory of Computational Physiology (LCP)

- Collaborating with Dr. Leo Anthony Celi from MIT's LCP on clinical multimodal discordance awareness and evaluation on MIMIC-IV dataset. Research is ongoing as a final capstone project for Harvard masters program.

Biomedical-KG: Knowledge-Graph Augmentation for Single-Cell Perturbation Prediction — MIT (Cross-Registered)

- Developed a model-agnostic framework with scGPT transcriptomic embeddings from Tahoe100M atlas to audit KG-bias (PrimeKG++) in single-cell drug perturbation predictions.
- Demonstrated that KG augmentation yields statistically significant improvements along with causal analysis on heavily-weighted pathways.

Computer Vision: AutoREPA Representation Learning for Diffusion Models — MIT (Cross Registered)

- Engineered a cosine annealing schedule that outperforms standard REPA alignment strategies across FID, KID, and IS metrics in 250k-step training regime.
- Conducted a three-stage study on a SiT-B diffusion transformer to optimize scheduling and layer placement of representation alignment on ImageNet100.
- Evaluated early-stop projection schedules and Pareto trajectories, discovering that alignment pressure is front-loaded and significantly alters the optimization path between image fidelity and diversity.

Domain-Transfer: Pneumonia Detection from X-Ray Fine-tuning — Harvard

- Engineered transfer learning architecture utilizing the self-supervised RAD-Dino foundational model and Low-Rank Adaptation (LoRA) for domain adaptation for pediatric X-ray scans.
- Analyzed the model's cross-domain generalization capabilities, actively mitigating modality-specific challenges such as pre-training data leakage and target-set data drift.

XAI - Detecting and Reducing Hallucinations in Factual Question Answering w/ Recursion — HKUST

- Reduced hallucinations by 8% on TriviaQA with multi-turn recursive LLaMA-7B framework.
- Developed a prefix-based sensitivity tuning method for factual grounding, achieving improvements without reliance on external oracles.
- Conducted empirical evaluation of hallucination mitigation strategies, analyzing error patterns and scaling behavior across datasets.

Publications & Awards

- Co-Author in "Financial Considerations in Primary Healthcare after COVID-19", *Springer Nature*. Featured on WHO Collaborative Center of Primary Health Care (PHC).
- TPG Fellowship for MSc. Big Data Technology
- Northwestern Dean's List

Community & Leadership

- Directed Forces at Work Charity Photobook Exhibition, raising HKD \$100K for Médecins Sans Frontières
- Winner, ACS Hackathon (2018): proposed haptic feedback gloves for visually impaired; awarded HKD \$25K

Skills

- Programming: C++, Python, R, MATLAB, AMPL
- ML: TensorFlow, PyTorch, HuggingFace, JAX, Unsloth, OpenCV, LangChain, pandas, NumPy
- Graph: iGraph, Networkx, pcst_fast, leidenalg
- Database: SQL, Spark, Hadoop
- Performance Optimization: CUDA, MPI, Pthreads
- Biomedical computation: AlphaFold, ColabFold, ColabDesign, PyMol, PyRosetta
- Languages: Native in English, Mandarin, Cantonese; Beginner in German